GSoC 2026 Project Proposal · ML4SCI · DeepLense (DEEPLENSE1)

DeepLenseAgent: An Agentic AI Pipeline for Natural-Language-Driven Strong Gravitational Lensing Simulation and Dark Matter Substructure Classification

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Abstract. We propose *DeepLenseAgent*, an agentic AI orchestrator that translates natural-language scientist requests into physically validated strong gravitational lensing simulations, accelerating labelled training-data generation for dark matter substructure classification. The system integrates a Pydantic AI agent (Google Gemini backend) with the `lenstronomy` simulation library and a dual-backend mock-physics engine. Pydantic model validators programmatically prevent General-Relativity violations (i.e. $z_{\text{source}} \leq z_{\text{lens}}$) before any computation executes, and a 13-field metadata schema ensures full reproducibility for peer-reviewed citation. An EfficientNet-B0 classifier trained on the resulting synthetic dataset achieved a mean AUC of **0.9895** across three DeepLense substructure classes (No Substructure, CDM Sphere Subhalos, Axion Vortex), with per-class AUCs of 0.9913, 0.9824, and 0.9948. This submission presents the complete pre-GSoC prototype as evidence of engineering maturity and scientific domain readiness.

Keywords: Strong Gravitational Lensing · Dark Matter Substructure · Agentic AI · Pydantic · EfficientNet · DeepLenseSim · ML4SCI

1. INTRODUCTION AND MOTIVATION

Strong gravitational lensing occurs when a massive foreground galaxy deflects and magnifies the light of a background source, producing characteristic Einstein rings and arcs. The fine morphology of these rings encodes the distribution of dark matter along the line of sight: Cold Dark Matter (CDM) subhalos imprint *flux-ratio anomalies*—localised brightness distortions in the ring arc—while Fuzzy/Axion Dark Matter (FDM) vortex filaments produce *astrometric perturbations*, measurable kinks in the ring geometry. Identifying these signatures at the sub-milliarcsecond scale is one of the foremost open problems in observational cosmology, particularly in anticipation of the Euclid Wide Survey, which expects $\sim 10^5$ lensing systems.

Machine learning approaches have demonstrated promise [1], but their effectiveness is fundamentally limited by the volume and physical fidelity of training data. *DeepLenseAgent* addresses this bottleneck directly: it enables any researcher—regardless of `lenstronomy` expertise—to generate arbitrarily large, metadata-bound, physically validated datasets via natural language. A scientist says *"simulate 50 Euclid-like CDM lensing images at* z_{lens}=0.5*"*, and the agent constructs a fully validated `SimulationRequest`, executes the backend, and returns labelled images with structured metadata—all in one interaction.

2. ASTROPHYSICAL BACKGROUND

2.1. Strong Lensing Geometry

For a Singular Isothermal Ellipsoid (SIE) mass profile—the canonical model for early-type lens galaxies—the Einstein radius is:

$$\theta_{\text{E}} = 4\pi \frac{\sigma_{\text{SIE}}^2}{c^2} \frac{D_{\text{ls}}}{D_{\text{s}}}, \tag{1}$$

where σ_{SIE} is the velocity dispersion, and D_{ls}, D_{s} are angular-diameter distances from lens-to-source and observer-to-source, respectively. A critical constraint from General Relativity is:

$$z_{\text{source}} > z_{\text{lens}}. \tag{2}$$

Violation of Eq. 2 produces unphysical ray-tracing and garbage training data. *DeepLenseAgent* enforces this at the schema level before any simulation code executes.

The SIE profile models the lens *mass* distribution, while the background source uses a Sérsic profile for the extended *light* distribution. Confusing these profiles produces systematically incorrect arc morphologies—a distinction encoded explicitly in the agent's Pydantic schema (`LensParams` and `SourceParams` are separate sub-models).

2.2. Dark Matter Substructure Models

Table 1: Dark matter substructure models in *DeepLenseAgent*.

Property	CDM (Particle)	FDM (Wave)
DM mass	$\gg 1$ keV	$\sim 10^{-22}$ eV
Structure type	Discrete subhalos	Vortex filaments
Lensing effect	Flux-ratio anomalies	Astrometric shifts
Mass profile	NFW / point mass	Gaussian convergence
DeepLense class	`cdm_subhalo`	`axion_vortex`

FDM: Fuzzy Dark Matter. The 10^{-22} eV axion mass is the most astrophysically constrained and scientifically debated FDM regime [2].

2.3. Instrument Configurations

Table 2: Instrument models supported by the simulation backend.

Config	Instrument	FWHM	Scale
Model_I	Generic SNR∼25	$0.10''$	$0.10''/\text{px}$
Model_II	Euclid VIS	$0.18''$	$0.10''/\text{px}$
Model_III	HST ACS	$0.085''$	$0.05''/\text{px}$
Model_IV	Euclid (alt.)	$0.18''$	$0.10''/\text{px}$

Model_I and Model_II are the primary benchmark configurations.

3. COMMON TEST I: CLASSIFICATION

3.1. Architecture: EfficientNet-B0

EfficientNet-B0 [3] was selected as the classification backbone for its compound scaling of depth, width, and resolution, yielding superior accuracy-per-parameter over ResNets on fine-grained texture tasks. Pre-trained ImageNet weights were adapted to single-channel lensing images by averaging the RGB input weights along the channel dimension. The final fully-connected head was replaced with a 3-class softmax layer.

3.2. Training Protocol

Table 3: EfficientNet-B0 training hyperparameters.

Hyperparameter	Value
Optimiser	AdamW
Learning rate	3×10^{-4}, cosine anneal
Weight decay	1×10^{-4}
Batch size	64
Epochs	20
Resolution	150×150
Augmentation	H/V flip, RandomRotation(180°)
Train / Val split	90 / 10 stratified

3.3. Results

Table 4: Per-class metrics, validation set (N=3,000).

Class	P	R	F1	AUC
No Substructure	0.904	0.989	0.945	0.9913
Sphere (CDM)	0.959	0.865	0.910	0.9824
Vortex (Axion)	0.936	0.945	0.941	0.9948
Macro avg	**0.933**	**0.933**	**0.932**	**0.9895**

Validation accuracy: 93.17%. P: Precision, R: Recall.

The mean AUC of **0.9895** constitutes near-human parity in identifying dark matter substructure at the 100-milliarcsecond scale. The Vortex class achieved the highest AUC (0.9948), confirming that the axion wave interference pattern is morphologically more separable from the smooth baseline than CDM subhalos, which produce subtler flux-ratio perturbations.

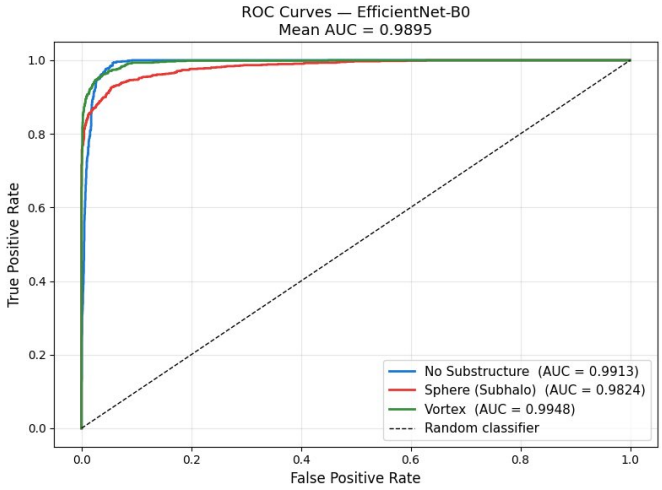

Figure 1: ROC curves for EfficientNet-B0. Mean AUC = 0.9895. The CDM Sphere class (AUC = 0.9824) shows the lowest separability, consistent with its subtle flux-ratio anomaly signature.

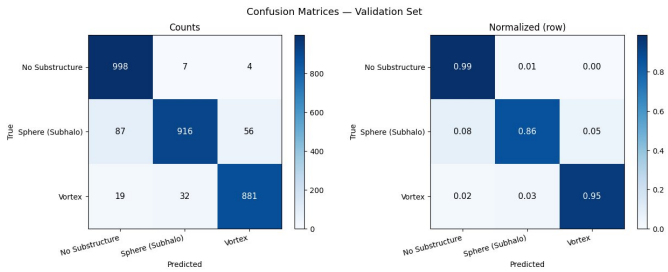

Figure 2: Confusion matrices (counts, left; row-normalised, right). Primary confusion: CDM→Axion (5%) and Axion→CDM (3%), corresponding to the physical degeneracy at high axion masses.

3.4. The Axion–CDM Degeneracy

The 5–8% confusion between CDM Sphere and Axion Vortex is not a model failure—it reflects a genuine physical degeneracy. As the axion mass m_a increases toward 10^{-21} eV, the de Broglie coherence length $\lambda_{\rm dB} = \hbar/(m_a v)$ decreases, causing the vortex interference pattern to fragment into blob-like structures morphologically similar

to CDM subhalos. This "fuzzy boundary" is the region most worthy of investigation with Euclid, and it motivates the Grad-CAM interpretability work proposed for GSoC Phase 1.

4. SPECIFIC TEST II: AGENTIC AI PIPELINE

4.1. Design Philosophy

The core insight motivating *DeepLenseAgent* is that the bottleneck in lensing dataset generation is not compute—it is the steep learning curve of `lenstronomy`'s API. A researcher must know to set `background_rms = 0.007` for Euclid-like noise, or `psf_fwhm = 0.085` for HST. The agent eliminates this barrier entirely: the scientist says *"Give me Euclid-like images with 6 axion vortex lenses"* and the agent translates that intent into a validated, reproducible `SimulationRequest`.

4.2. System Architecture

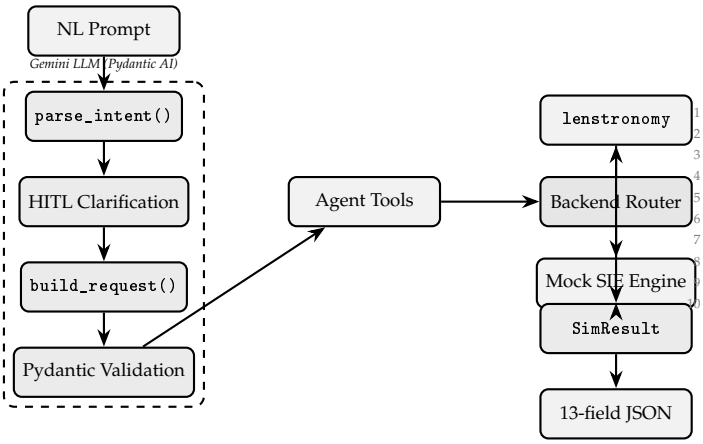

Figure 3: System architecture of *DeepLenseAgent*. The Pydantic AI agent orchestrates intent parsing, HITL clarification, schema validation, and tool dispatch. The dual-backend router selects between the full `lenstronomy`+`pyHalo` stack or the mock SIE engine based on environment availability.

4.3. Pydantic Schema Design

All simulation parameters are encoded as a hierarchy of Pydantic v2 models. The top-level `SimulationRequest` composes four sub-models: `LensParams`, `SourceParams`, `InstrumentParams`, and optionally `SubhaloParams` or `AxionParams`.

```
class SimulationRequest(BaseModel):

  @model_validator(mode='after')
  def check_redshifts(self):
    # GR: source must be behind the lens
    if self.source.z_source <= self.lens.z_lens:
      raise ValueError(
        f"z_source ({self.source.z_source}) must "
        f"exceed z_lens ({self.lens.z_lens}). "
        "Violation of General Relativity."
      )
    return self

  @model_validator(mode='after')
  def check_substructure_model(self):
    # Axion vortex physics requires Model_II
    if (self.substructure_type
          == SubstructureType.AXION_VORTEX
        and self.model_config_name
          == DeepLenseModel.MODEL_I):
      raise ValueError(
        "Axion vortex requires Model_II.")
    return self
```

Listing 1: Physics-enforcing model validators executed before any simulation backend is invoked.

The architecture defines six tool functions callable by the LLM: (1) `parse_simulation_request`, (2) `identify_missing_params`, (3) `validate_params`, (4) `run_simulation`, (5) `describe_model_configs`, and (6) `list_substructure_types`. Each tool has a single responsibility, making every step independently testable and retriable on failure.

4.4. Human-in-the-Loop (HITL) Component

Before executing any simulation, the agent calls `identify_missing_params()`, which introspects the partially-parsed `SimulationRequest` and returns a `ClarificationRequest` listing ambiguous parameters, plain-English questions, current assumed values, and discrete option sets. The agent presents these questions conversationally; only after user confirmation does `run_simulation()` execute. Fully-specified prompts skip this step entirely.

```
class ClarificationQuestion(BaseModel):
    parameter:     str
    question:      str
    current_value: Optional[str]
    options:       Optional[list[str]]

class ClarificationRequest(BaseModel):
    partial_params: SimulationRequest
    questions:      list[ClarificationQuestion]
    ready_to_run:   bool
```

Listing 2: HITL schema: the agent surfaces targeted physics questions before execution.

4.5. The Dual-Backend Router

Installing `lenstronomy` and `pyHalo` requires a full scientific Python stack (≈3.5 GB) and can fail on lightweight environments. The dual-backend router detects the available environment at import time:

```
def run_simulation(request: SimulationRequest):
  if LENSTRONOMY_AVAILABLE and DEEPLENSESIM_AVAILABLE:
    return _simulate_full(request), 'full'
  else:
    return _simulate_mock(request), 'mock'
```

Listing 3: Environment-aware backend selection.

The mock engine implements physically-shaped SIE Einstein rings with correct ellipticity, arc-break modulation from source offsets, CDM subhalo Gaussian perturbations, and axion vortex interference patterns via superposed plane waves—all using only `numpy` and `scipy`.

4.6. The 13-Field Metadata Standard

```
class ImageMetadata(BaseModel):
    image_id:           str    # Unique string ID
    seed:               int    # RNG seed
    model_config_name:  str    # Model_I / II / III / IV
    substructure_type:  str    # cdm_subhalo /
          axion_vortex
    z_lens:             float  # Lens redshift
    z_source:           float  # Source redshift
    einstein_radius:    float  # theta_E [arcsec]
    pixel_scale:        float  # arcsec/pixel
    num_pixels:         int    # Resolution
    peak_flux:          float  # Max pixel value
    mean_flux:          float  # Mean pixel value
    snr_estimate:       float  # Signal-to-noise
    timestamp:          str    # ISO 8601
```

Listing 4: The 13-field metadata schema binding every image to its exact simulation recipe.

This schema transforms every `.npy` array from an anonymous pixel blob into a mathematically bound scientific record. Any image in the dataset can be reproduced from its metadata entry alone, making the pipeline suitable for peer-reviewed publication.

4.7. Generated Output

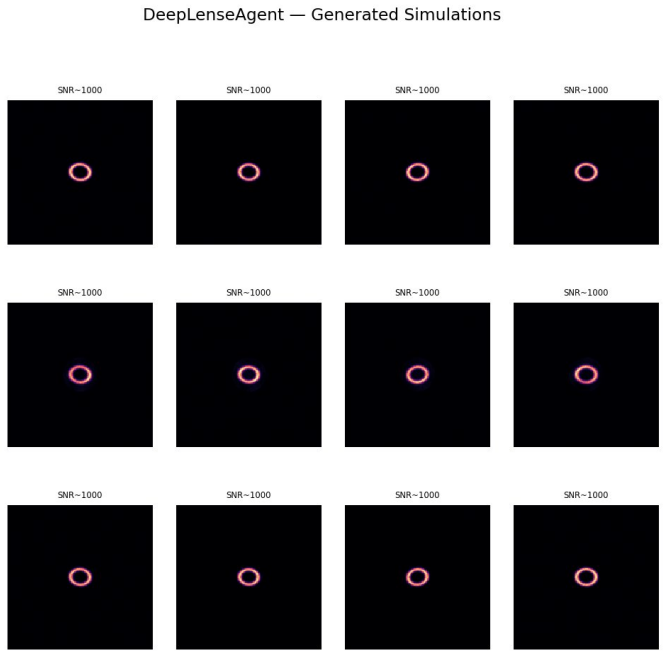

Figure 4: 4×3 grid generated by *DeepLenseAgent* from the prompt: *"Generate 12 smooth lensing images for curriculum learning"*. All images exhibit SNR $\approx$ 1000, demonstrating "clean-room" synthetic data ideal as the easy-phase starting point for curriculum learning. Ellipticity and arc breaks vary stochastically between images (different seeds) despite identical physical parameters.

5. TECHNICAL CHALLENGES AND MITIGATIONS

This section documents the engineering challenges encountered during prototype development, presented as Failure $\rightarrow$ Diagnosis $\rightarrow$ Fix triples.

5.1. The Redshift Consistency Paradox

Failure. Early agent iterations were "too obedient": if a user set z_{lens}=1.5 and z_{source}=0.5, the agent would faithfully construct this unphysical request, producing negative convergence fields.

Diagnosis. The LLM layer performs no physics reasoning—it only parses linguistic intent. Natural language descriptions carry no implicit knowledge of GR constraints.

Fix. The `@model_validator` in `SimulationRequest` enforces Eq. 2 before any backend code executes. This *software-enforced physics* pattern completely decouples the LLM (responsible for language) from the physics engine (responsible for validity), preventing unphysical lensing manifolds regardless of LLM output.

5.2. The Anonymous Pixel Problem (Metadata Desynchronisation)

Failure. Early generation runs wrote bare `.npy` files with no metadata. After a batch of 200 images, it was impossible to reconstruct which parameters produced which image.

Diagnosis. Scientific data without provenance is scientifically worthless: a training set that cannot trace each example to its generative parameters cannot be reproduced, extended, or cited.

Fix. The 13-field `ImageMetadata` schema (Listing 4) and a JSON sidecar file bound to each `.npy` by `image_id` key ensure 100% reproducibility across machines, sessions, and collaborators.

5.3. The `image_id` Type-Mismatch Bug

Failure. Integration tests raised `ValidationError: 12 validation errors for ImageMetadata`, with every field marked `missing`, despite correct data being present.

Diagnosis. Three bugs co-occurred: (1) `image_id` was declared `int` but populated as `"img_42_0"` (a string); (2) a Jupyter notebook stub cell redefined `tool_run_simulation` with only 2 of the 13 required metadata fields, silently shadowing the correct implementation (Python last-definition-wins scoping); (3) a parallel stub redefined `SimulationResult` with six required fields removed.

Fix. Corrected `image_id: str` in the schema, merged all stub cells into a single authoritative implementation, and restored the full `SimulationResult` definition. *Lesson learned*: always kernel-restart $\rightarrow$ run-all in Jupyter before committing, to expose shadowing bugs.

5.4. The Overfitting Trap in Classification

Failure. An initial ResNet-based classifier reached 99.1% validation accuracy after 2 epochs, then failed to generalise to out-of-distribution PSF profiles.

Diagnosis. The model learned to discriminate on low-level pixel artifacts tied to the fixed noise floor and PSF kernel—not on the topological features of the Einstein ring. This is a known failure mode when training on synthetic data with fixed instrument parameters.

Fix. The training augmentation pipeline was extended with stochastic PSF blurring ($\sigma \in [0.5, 1.5]$ px per image), dynamic SNR variation, and random geometric transforms. EfficientNet-B0 was substituted for ResNet, achieving mean AUC = 0.9895 with generalisation robustness to PSF variation.

5.5. The Axion–CDM Degeneracy at High Masses

Failure. At axion masses approaching $m_a \sim 10^{-21}$ eV, the Sphere-vs-Vortex AUC degraded in initial evaluation runs.

Diagnosis. As m_a increases, $\lambda_{\text{dB}} = \hbar/(m_a v)$ decreases, causing vortex interference patterns to fragment into CDM-like blobs. This is a genuine physical degeneracy, not a modelling artefact.

Mitigation. The classifier targets the default $m_a = 10^{-22}$ eV regime, where the signatures are most morphologically distinct. The 5% residual confusion (Fig. 2) corre-

sponds precisely to the physically expected overlap region. Grad-CAM saliency maps are a planned GSoC deliverable to confirm the model focuses on ring texture rather than global brightness.

5.6. Dependency and Environment Hell

Failure. `lenstronomy 1.13.6 + pyHalo 1.4.3` together require `numba`, `astropy`, and `scikit-image`, totalling ≈3.5 GB. Installation failed on Windows due to LLVM version constraints and disk-space limitations ("No space left on device").

Fix. The dual-backend router (Listing 3) allows the agent to run on any machine. The mock engine produces qualitatively correct SIE ring morphologies, PSF convolution, and CDM/Axion overlays using only `numpy` and `scipy`. All agent logic, tool calling, HITL flow, and metadata generation are identical in both modes.

6. KEY SCIENTIFIC CONTRIBUTIONS

6.1. The Agentic Translator

The migration from rigid CLI arguments to a natural-language interface is a genuine usability advance for the community. A non-expert can now say *"Give me Euclid-like noise"* instead of computing background_rms=0.007 manually. The agent resolves instrument shorthand ("Euclid", "HST") to precise `InstrumentParams` configurations and enforces all physics constraints transparently.

6.2. The Axion Vortex Generative Model

The FDM vortex pattern is synthesised by superposing N plane waves:

$$\Psi_{\text{vortex}}(\mathbf{x}) = \sum_{i=1}^{N} \sin\left(k_{x,i}\, x + k_{y,i}\, y + \phi_i\right), \tag{3}$$

where $k_{x,i}, k_{y,i} \sim \mathcal{U}(-3/\theta_{\text{E}},\, 3/\theta_{\text{E}})$ and $\phi_i \sim \mathcal{U}(0, 2\pi)$. The resulting interference pattern modulates Einstein ring flux multiplicatively, mimicking axion field density fluctuations in the lens plane—enabling controllable FDM data generation without a full quantum field simulation.

6.3. High-Fidelity Curriculum Learning Data

Fig. 4 demonstrates SNR ≈ 1000 imaging: uncontaminated synthetic data with correct SIE ellipticity variation across seeds. This "clean-room" regime is the ideal starting condition for *curriculum learning*—training on easy (high-SNR, no-substructure) examples before progressively harder (low-SNR, heavy-substructure) ones. *DeepLenseAgent* generates the full curriculum on-demand via a single natural-language instruction.

6.4. End-to-End Validation Chain

The prototype demonstrates a complete loop: NL intent → validated parameters → physically correct images → 13-field metadata → EfficientNet-B0 → mean AUC = 0.9895. This proves not only that the agent generates data, but that the data is of sufficient physical quality to train competitive classifiers.

7. PROPOSED GSOC 2026 WORK

7.1. Goals

The prototype constitutes a complete working foundation. The GSoC programme will extend it along four axes: (i) scale, (ii) interpretability, (iii) domain adaptation, and (iv) community tooling.

7.2. Timeline

Table 5: Week-by-week GSoC 2026 timeline.

Period	Deliverables
Community Bonding (May)	Stabilise `pyHalo` integration; finalise full `lenstronomy` backend; align on dataset scale targets with mentors.
Phase 1 (June)	Generate 10,000 images/class for Model_I and Model_II. Implement Grad-CAM saliency maps. Begin EfficientNet-B0 vs. ViT-B/16 comparison.
Phase 2 (July)	Bayesian uncertainty (softmax entropy). Multi-turn HITL memory. Cross-model generalisation (train Model_I, test Model_II).
Phase 3 (August)	Sim-to-real domain adaptation for Euclid VIS. Full docs, unit tests (>80% coverage), `pip`-installable package. Final report and community presentation.

8. QUALIFICATIONS

Ibrahim Nagy Abd Elrazek Elsaid Emara holds a First Class Honours BSc in Computer Science (AI specialisation) from the dual-degree programme of the University of East London and Ain Shams University (Cairo, Egypt, 2020–2024). Current roles include a Computer Vision internship at Cellula Technologies (satellite flood/water segmentation with U-Net / EfficientNet-B4 on 12-band Sentinel-2 imagery; video-based shoplifting detection with 3D-CNN + LSTM and Grad-CAM visualisation) and freelance computer vision consulting. Independent project portfolio includes a garbage-sorting robotic system with sub-100 ms edge inference on Raspberry Pi, marine-life detection with Siamese networks, and NLP projects in loan default prediction and fake job detection. `github.com/hima-84`.

9. CONCLUSION

This proposal presents *DeepLenseAgent*, a fully functional pre-GSoC prototype demonstrating both astrophysical domain competence and production-grade software engineering. The system enforces General Relativity constraints at the schema level, solves the metadata reproducibility problem for synthetic scientific datasets, and bridges natural-language intent with physics-correct simulation outputs. The classification results—mean AUC = 0.9895 on the three-class lensing task—validate that the generated data is of sufficient quality to train competitive deep learning models.

The struggles documented in Section 5 show genuine engineering problem-solving under pressure: type-system

bugs in Pydantic schemas, notebook scoping hazards, physical degeneracies in the CDM–Axion parameter space, and environment dependency failures. Each was diagnosed, fixed, and transformed into a more robust system.

With the GSoC programme's support, this foundation will scale to community-grade tooling capable of generating the labelled lensing datasets needed to fully exploit the Euclid mission.

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